SIMATS ENGINEERING (SAVEETHA SCHOOL OF ENGINEERING)

Department of Computer Science and Engineering

ASSESSMENT TOOL 2 – Case Study

Course Code:	**ITA0517**	**Course Title:**	**Computer Vision**
CO Assessed:	**CO3 – Precision (accurate feature extraction & analysis) (BL3)**	**Assessment:**	**Assessment Tool 2 – Case Study**
Weightage:	**40% of CIA**	**Total Marks:**	**100**
CO3 Statement:	*Analyze and extract image features using detection and matching techniques for effective image representation and comparison. (BL4)*		
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Section / Batch:	CSE-AI/2024-2028	**Date:**	10-08-2026

Code of Conduct

I Malli Chandana/192472250(Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment. I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the Student:Malli Chandana

Criteria	Weigh tage	Level 4 – Exemplary	Level 3 – Proficient	Level 2 – Developing	Level 1 – Beginning	Marks
Understanding of Case	**25**	**Thorough understanding; clearly identifies all key issues**	**Good understanding with most issues identified**	**Basic understanding; some issues missed**	**Poor understanding; problem not identified correctly**	
Application of Concepts	**25**	**Effectively applies relevant concepts to analyze the case deeply**	**Applies appropriate concepts with minor gaps**	**Limited application of concepts**	**Fails to apply relevant concepts**	
Analysis	**20**	**In-depth analysis with strong reasoning and insights**	**Good analysis with logical reasoning**	**Basic analysis with limited depth**	**Weak or no analysis; lacks reasoning**	
Solution Design	**20**	**Innovative, feasible solutions with strong justification**	**Logical solutions with adequate justification**	**Basic solutions with weak justification**	**Incorrect or no solution provided**	
Clarity	**10**	**Clear, well-structured, and easy to understand**	**Organized and understandable presentation**	**Limited clarity and structure**	**Poor clarity; difficult to understand**	

CASE STUDY: COMBINED FEATURE DETECTION AND MATCHING

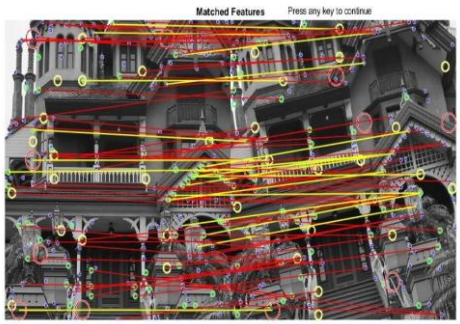
1. Introduction

Feature detection and feature matching are important techniques in Computer Vision. They are used to identify important points in an image and determine whether those points correspond to the same locations or objects in another image.

A typical feature-based vision system consists of three major stages:

1. **Feature Detection** – identifying distinctive points or regions in an image.
2. **Feature Description** – representing the detected features using numerical descriptors.
3. **Feature Matching** – comparing descriptors from two images to find corresponding features.

In this case, the system successfully detects features from two images, but the matching stage produces incorrect or very few matches. The main reason is the selection of an inappropriate feature descriptor for the detected features.This case study analyzes the problem and explains how selecting an appropriate descriptor can significantly improve matching performance.



2. Problem Statement

A computer vision system is developed to detect and match features between two images. The feature detector successfully identifies important points such as corners, edges, and textured regions.

However, during the matching stage, the system produces:

- Very few correct matches
- A large number of incorrect matches
- Unstable matching results
- Poor correspondence between the two images
- Difficulty matching features under changes in scale, rotation, illumination, or viewpoint

The investigation shows that the main problem is not necessarily the feature detector. Instead, an unsuitable feature descriptor has been selected.

Therefore, the objective is to analyze the descriptor-selection problem and identify an appropriate descriptor that improves the accuracy and reliability of feature matching.

3. Objectives

The main objectives of this case study are:

1. To understand the relationship between feature detection, feature description, and feature matching.
2. To identify why correct feature detection can still result in poor matching.
3. To analyze the effect of inappropriate descriptor selection.
4. To compare commonly used feature descriptors such as SIFT, SURF, ORB, and BRIEF.
5. To select a suitable descriptor according to the characteristics of the images.
6. To improve matching accuracy and reduce false matches.
7. To understand the importance of distance metrics and matching algorithms.
8. To design a reliable feature detection and matching pipeline.

4. Understanding Feature Detection and Matching

Feature detection identifies distinctive locations in an image. These locations are generally called **keypoints** or **features**.

Examples of detectable features include:

- Corners
- Edges
- Blobs
- Highly textured regions
- Distinctive object points

Popular feature detectors include:

- Harris Corner Detector
- FAST
- SIFT
- SURF
- ORB

After detecting a feature, the system needs to describe the local image region around that feature. This representation is called a **feature descriptor**.The descriptors from two images are then compared to determine which features correspond to each other.

The general pipeline is:

Input Images → Feature Detection → Feature Description → Descriptor Matching → Filtering → Final Correspondences

5. Case Description

Consider an image-matching application used to identify the same object in two different photographs.For example, Image 1 contains a book photographed from the front, while Image 2 contains the same book photographed from a slightly different angle.The feature detector correctly identifies several keypoints around:

- Text regions
- Corners
- Logo areas
- Object boundaries
- Textured regions

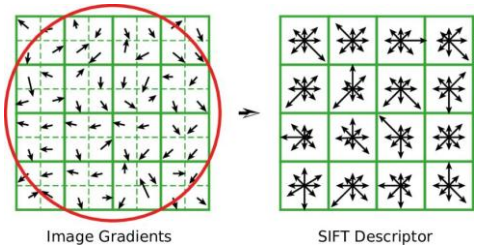
However, the system uses an unsuitable descriptor. As a result, the descriptor vectors do not represent the local features reliably.

During matching, the system may therefore:

- Match unrelated points.
- Fail to match corresponding points.
- Produce many outliers.
- Generate unstable results.

This demonstrates an important principle:

Good feature detection does not guarantee good feature matching: The descriptor must represent the detected feature in a way that allows corresponding regions to be recognized reliably.



6. Why Descriptor Selection Is Important

A feature detector answers the question:

"Where is an interesting point?"

A descriptor answers the question:

"What does the region around this point look like?"

The matching algorithm then compares these descriptions.If the descriptor is not suitable for the image conditions, two visually corresponding points may have very different descriptors. Similarly, two unrelated points may accidentally have similar descriptors.Therefore, descriptor

selection has a direct influence on matching performance.For example, consider two images of the same object captured after rotating the camera.If the selected descriptor is not sufficiently rotation-invariant, the same physical feature may produce different descriptor values in the two images. The matcher may then fail to identify them as corresponding points.

7. Common Feature Descriptors

7.1 SIFT

SIFT stands for **Scale-Invariant Feature Transform**.

SIFT generates descriptors that are relatively robust to:

- Scale changes
- Rotation
- Moderate illumination changes
- Some viewpoint changes

SIFT descriptors are commonly used when accuracy and robustness are more important than computational speed.

Advantages

- Strong scale invariance
- Strong rotation invariance
- Good matching accuracy
- Effective for images containing detailed textures

Disadvantages

- Computationally more expensive than binary descriptors
- Larger descriptor representation
- May be slower for real-time applications

7.2 SURF:SURF stands for **Speeded-Up Robust Features**.It was designed to provide robust feature detection and description while improving computational efficiency compared with traditional approaches.

Advantages

- Robust to scale and rotation changes
- Good descriptor performance
- Faster than some traditional approaches

Disadvantages

- More computationally expensive than lightweight binary methods
- Availability and usage may depend on the computer vision library and implementation

7.3 ORB

ORB stands for **Oriented FAST and Rotated BRIEF**.

ORB combines the FAST keypoint detector with a rotated version of the BRIEF descriptor.

ORB is particularly useful for applications where computational efficiency is important.

Advantages

- Fast
- Computationally efficient
- Suitable for real-time applications
- Uses binary descriptors
- Requires less memory than many floating-point descriptors

Disadvantages

- Generally less robust than SIFT under large scale or viewpoint changes
- May perform poorly on images with significant appearance changes
- **7.4 BRIEF:**BRIEF stands for **Binary Robust Independent Elementary Features**.It represents local image regions using a binary string generated from intensity comparisons.

Advantages

- Very fast
- Compact representation
- Efficient matching using Hamming distance

Disadvantages

- Basic BRIEF is not inherently rotation invariant
- Performance can decrease when the image undergoes significant rotation

9. Comparison of Feature Descriptors

Descriptor	Descriptor Type	Rotation Robustness	Scale Robustness	Speed	Typical Application
SIFT	Floating-point	High	High	Moderate	Accurate image matching
SURF	Floating-point	High	High	Moderate	Object recognition
ORB	Binary	Good	Moderate	Very High	Real-time vision
BRIEF	Binary	Low	Low	Very High	Fast feature matching

BRIEF Binary Low Low Very High Fast feature matching

The appropriate descriptor depends on the requirements of the application.

For example:

- **High accuracy:** SIFT
- **Real-time processing:** ORB
- **Low computational cost:** ORB/BRIEF
- **Large scale changes:** SIFT
- **Significant rotation:** SIFT or ORB

9. Analysis of the Problem

The system in this case successfully detects features but fails during matching. This indicates that the feature detection stage is not necessarily the main problem.The following factors may be responsible.

9.1 Incorrect Descriptor Selection: The selected descriptor may not be suitable for the image transformation.For example, if images contain significant rotation and a descriptor with poor rotation invariance is used, corresponding features may not match correctly.

9.2 Descriptor and Matcher Incompatibility

Different descriptor types require appropriate distance metrics.

For example:

- Floating-point descriptors such as SIFT are commonly compared using Euclidean distance.
- Binary descriptors such as ORB and BRIEF are commonly compared using Hamming distance.

Using an inappropriate distance metric can result in poor matching.

9.3 Scale Changes: If the object appears significantly larger or smaller in another image, a descriptor with weak scale robustness may fail to represent corresponding features consistently.

9.4 Illumination Changes: Changes in brightness, shadows, or exposure can modify the appearance of local regions.A descriptor that is sensitive to illumination changes may generate substantially different representations for the same physical feature.

9.5 Viewpoint Changes: When an object is photographed from different viewpoints, the appearance of local regions can change.Large viewpoint changes can cause some descriptors to become unreliable, resulting in incorrect matches or missing correspondences.

9.6 Repetitive Patterns: Images containing repeated patterns can create ambiguous matchesFor example, a building containing many identical windows may produce several visually similar features. The matcher may incorrectly associate one window with another.

10. Incorrect Descriptor Selection – Example

Suppose two images of the same object are captured.

Image 1

The object is photographed normally.

Image 2

The same object is rotated by approximately 90 degrees.The feature detector identifies the same physical corners in both images.However, suppose a rotation-sensitive descriptor is used.The descriptor generated for a corner in Image 1 may differ significantly from the descriptor generated for the corresponding corner in Image 2.

As a result:

Correct Feature Detection → Incorrect Descriptor Representation → Poor Matching

This shows that feature detection alone is insufficient.

11. Effect of Appropriate Descriptor Selection:

Selecting an appropriate descriptor improves the matching process in several ways.

11.1 Improved Correspondence:A good descriptor generates similar representations for corresponding features.Therefore, the matcher can identify the correct pairs more reliably.

11.2 Reduced False Matches:A discriminative descriptor makes unrelated features easier to distinguish.This reduces the number of incorrect correspondences.

11.3 Better Robustness: Appropriate descriptors can provide robustness against:

- Rotation
- Scale changes
- Illumination changes
- Moderate viewpoint changes
- Noise

11.4 Improved Matching Accuracy: When the descriptor accurately represents the local image region, the percentage of correct matches increases.

11.5 Better Object Recognition: Reliable feature matching can improve higher-level computer vision tasks such as:

- Object recognition
- Image stitching
- Panorama generation
- Augmented reality
- Visual localization
- 3D reconstruction
- Image registration

- Robot navigation

12. Proposed Solution: The proposed solution is to select the descriptor according to the image characteristics and application requirements.

The improved pipeline is:

Input Images

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Preprocessing

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Feature Detection

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Appropriate Feature Descriptor

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Descriptor Matching

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Ratio Test / Distance Filtering

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Outlier Removal

↓

RANSAC

↓

Final Correct Matches

13. Recommended Descriptor Selection

For a general-purpose image matching system, **SIFT** can be selected when robustness and matching accuracy are the primary requirements.For real-time systems where processing speed is important, **ORB** is a suitable alternative.

Recommended approach

If the images have:

- Large scale differences → Use SIFT
- Significant rotation → Use SIFT or ORB
- Real-time requirements → Use ORB
- Limited computational resources → Use ORB

- High accuracy requirements → Prefer SIFT

Therefore, descriptor selection should be based on the expected transformations and computational requirements of the application.

14. Matching Techniques

After generating descriptors, a matching algorithm is required.

14.1 Brute-Force Matcher:A Brute-Force matcher compares descriptors from one image with descriptors from another image.For SIFT-type floating-point descriptors, a suitable distance is generally Euclidean distance.For ORB and other binary descriptors, Hamming distance is commonly used.

14.2 K-Nearest Neighbour Matching:KNN matching finds the closest candidate descriptors for each feature.It can be combined with a ratio test to reject ambiguous matches.

14.3 Lowe's Ratio Test:The ratio test compares the distance of the best match with the distance of the second-best match.If the best match is significantly better than the second-best match, the correspondence is considered more reliable.This helps remove ambiguous matches.

14.4 RANSAC:RANSAC stands for **Random Sample Consensus**.

It can be used to estimate a geometric transformation while rejecting incorrect matches.

RANSAC is especially useful when a matching result contains both:

- Inliers – geometrically consistent matches
- Outliers – incorrect matches

Therefore, using descriptor matching together with geometric verification can substantially improve the reliability of the final result.

15. Improved System Design

The improved system should follow these steps:

Step 1: Read the images: Load the two images that need to be matched.

Step 2: Convert to grayscale: Convert the images into grayscale if the selected feature method works on grayscale intensity information.

Step 3: Detect features: Use an appropriate detector such as SIFT or ORB.

Step 4: Generate descriptors: Generate descriptors for every detected keypoint.

Step 5: Select an appropriate matcher: Choose the distance metric according to the descriptor type.

Step 6: Perform descriptor matching: Match the descriptors between the two images.

Step 7: Apply filtering: Use KNN matching and a ratio test to eliminate weak or ambiguous matches.

Step 8: Remove geometric outliers: Use RANSAC to identify geometrically consistent matches.

Step 9: Display the results: Draw the final reliable matches between the two images.

16. Example Scenario

Consider a mobile application that identifies a product from a photograph.A user captures a product image from a slightly different angle.The system detects many keypoints successfully.However, the original implementation uses an inappropriate descriptor. Consequently, only a small number of features match correctly.After replacing the descriptor with a more suitable one and using an appropriate matching distance, the number of correct correspondences increases.The system can then reliably identify the product.This example demonstrates that improving the descriptor can be more important than simply increasing the number of detected features.

17. Before and After Analysis

Parameter	**Incorrect Descriptor**	**Appropriate Descriptor**
Feature detection	Good	Good
Descriptor representation	Poor	Reliable
Correct matches	Low	High
False matches	High	Low
Rotation robustness	Poor	Better
Scale robustness	Poor	Better
Matching stability	Low	High

18. Evaluation Parameters

The performance of the system can be evaluated using:

Number of detected keypoints: Measures how many features were identified.

Number of matches: Measures how many descriptor correspondences were found.

Number of correct matches: Measures the useful correspondences after filtering.

Number of outliers: Measures incorrect feature correspondences.

Matching accuracy:A simple evaluation can be based on the proportion of correct matches among the accepted matches.

Processing time: Important for real-time applications.A good feature matching system should achieve a suitable balance between accuracy and computational efficiency.

19. Real-World Applications

Appropriate feature descriptors are important in many real-world applications.

19.1 Image Stitching: Features from overlapping images are matched to create panoramic images.

19.2 Augmented Reality: Real-world objects and surfaces can be recognized by matching visual features.

19.3 Object Recognition: An object can be identified by comparing its features with a reference image.

19.4 Robotics: Robots can use visual features for localization and navigation.

19.5 Image Registration: Images from different sources can be aligned using corresponding feature points.

19.6 3D Reconstruction: Feature correspondences between multiple images are used to estimate 3D structure.

19.7 Visual Search:A photographed object can be compared with a database of images to find similar objects.

20. Advantages of the Proposed Solution

The proposed approach provides several advantages:

1. Improves the number of correct feature matches.
2. Reduces false correspondences.
3. Improves robustness to image transformations.
4. Makes the matching process more reliable.
5. Allows the descriptor to be selected according to application requirements.
6. Supports both high-accuracy and real-time applications.
7. Can be combined with ratio testing and RANSAC for additional reliability.
8. Improves the performance of higher-level computer vision systems.

21. Limitations

Although appropriate descriptor selection improves performance, some limitations remain.

1. Large viewpoint changes can still cause matching failures.
2. Extreme illumination changes may affect descriptors.
3. Highly repetitive patterns can create ambiguous matches.
4. Complex images may require additional filtering.
5. High-accuracy descriptors can require more computational resources.
6. No single descriptor performs best for every type of image.

Therefore, descriptor selection should always consider the characteristics of the application.

22. Final Analysis

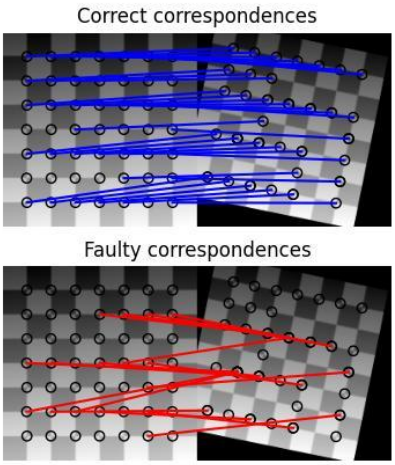
The case demonstrates that feature detection and feature matching are two different but interconnected processes.A system may detect features correctly but still fail to match them because the descriptors do not represent the local image regions appropriately.

The key issue is therefore:**Correct feature detection ≠ correct feature matching.**

A suitable descriptor should be selected based on factors such as:

- Rotation
- Scale
- Illumination
- Viewpoint
- Computational requirements
- Image texture
- Required accuracy

SIFT provides strong robustness and is suitable when matching accuracy is important. ORB provides a faster and computationally efficient alternative for real-time applications.Furthermore, descriptor matching should be combined with suitable distance metrics, ratio testing, and geometric verification such as RANSAC.



23. Conclusion

Feature detection and feature matching form an important part of many Computer Vision systems. The success of a matching system depends not only on detecting good keypoints but also on describing those keypoints appropriately.In the given case, the system detects features correctly but fails during matching because of incorrect descriptor selection. The problem can be solved by choosing a descriptor according to the characteristics of the images and the requirements of the application.SIFT is a suitable choice when robustness to scale and rotation and high matching accuracy are important.

Feature Detection → Appropriate Description → Correct Distance Metric → Matching → Ratio Test → RANSAC → Reliable Correspondences

Selecting the appropriate feature descriptor significantly improves matching accuracy, reduces false matches, and makes the overall computer vision system more reliable.

24. Key Takeaways

- Feature detection identifies important image points.
- Feature descriptors represent the local appearance around those points.
- Matching compares descriptors between images.
- Incorrect descriptor selection can cause matching failure even when feature detection is successful.
- SIFT is robust and suitable for accuracy-oriented applications.
- ORB is fast and suitable for real-time applications.
- Binary descriptors require appropriate binary distance measures such as Hamming distance.
- Floating-point descriptors can use distance measures such as Euclidean distance.
- Ratio testing helps remove ambiguous matches.
- RANSAC helps eliminate geometrically inconsistent matches.
- Appropriate descriptor selection is essential for reliable computer vision systems.

25. References

1. OpenCV Documentation – Feature Detection and Description.
2. D. G. Lowe, “Distinctive Image Features from Scale-Invariant Keypoints.”
3. E. Rublee et al., “ORB: An Efficient Alternative to SIFT or SURF.”
4. Computer Vision: Algorithms and Applications – Richard Szeliski.
5. OpenCV tutorials on feature matching and homography.